

Hadi Reisizadeh

Postdoctoral Associate, Optimization for AI Lab, University of Minnesota

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Summary

Research Scientist specializing in **LLM safety, optimization, and trustworthy AI**. Discovered that state-of-the-art unlearning methods fail to erase knowledge under probabilistic decoding. Developed **BLUR**, a bi-level optimization framework that achieves up to 19% higher unlearning efficiency while preserving model utility on LLaMA-2-7B and Zephyr-7B-Beta. Proposed an edge-preserving chain-of-thought condensation method that delivers lossless reasoning accuracy at 34% lower training cost across Qwen and LLaMA families. Built provably convergent distributed training algorithms for noisy, and time-varying networks. Seeking to advance AI systems that are not only capable but whose behavior can be verified, corrected, and trusted.

Technical Skills

Core Expertise	LLMs, optimization, machine unlearning, efficient training, and safe AI
Programming Languages	Python, MATLAB, R, C/C++
Machine Learning	PyTorch, HuggingFace Transformers, TensorFlow, Scikit-learn
Optimization	CPLEX, Gurobi

Work Experience

University of Minnesota, Minneapolis, MN

POSTDOCTORAL ASSOCIATE, OPTIMIZATION FOR AI LAB

09/2024 – PRESENT

RESEARCH ASSISTANT, INFORMATION PROCESSING GROUP

09/2017 – 09/2024

• LLM Unlearning and Model Safety.

- Investigated the fundamental limits of unlearning in LLMs, showing that forgotten knowledge can re-emerge under probabilistic decoding. Developed a **meta-evaluation framework** to quantify residual knowledge leakage and expose limitations in current unlearning methods across *TOFU*, *MUSE*, and *WMDP* [\[paper\]](#).
- Proposed a **bi-level optimization framework (BLUR)** for LLM unlearning that prioritizes the forget objective over the retain loss, achieving up to **19%** higher unlearning efficiency and **11.7%** higher model utility on *MUSE-News* with the **LLaMA-2-7B** model, and **16.6%** higher efficiency on *WMDP* with equivalent utility using the **Zephyr-7B-Beta** model [\[paper\]](#), [\[code\]](#).

• Efficient Reasoning Condensation.

Proposed an **edge-preserving condensation framework** for chain-of-thought reasoning training that retains only the initial and final reasoning stages, enabling supervised models to achieve **lossless reasoning accuracy** while reducing training time by **over 34%** across multiple **LLM families (Qwen, LLaMA)** and reasoning benchmarks [\[paper\]](#), [\[code\]](#).

• Distributed Training.

Designed a **two-time-scale optimization algorithm** for communication-efficient distributed training over noisy, time-varying networks, achieving **provable convergence guarantees** in both convex and non-convex settings [\[paper 1\]](#), [\[paper 2\]](#), [\[paper 3\]](#).

Seagate Technology, Shakopee, MN

05/2023 – 04/2024

RESEARCH INTERN, DATA TRUST RESEARCH TEAM

• Differentially Private Distributed Training.

Developed an **accelerated privacy-preserving distributed training method**, achieving faster convergence without reducing privacy guarantees.

Education

Ph.D. & M.Sc. in Electrical and Computer Engineering (*Minor in Computer Science*)

University of Minnesota, Twin Cities, Minneapolis, MN

09/2017 – 09/2024

B.Sc. in Electrical and Computer Engineering

Isfahan University of Technology, Isfahan, Iran

09/2012 – 05/2017

Selected Publications

- H. Reisizadeh**, J. Ruan, Y. Chen, S. Pal, S. Liu, and M. Hong, “*Leak@k: Unlearning Does Not Make LLMs Forget Under Probabilistic Decoding*,” ICML, 2026.
- H. Reisizadeh**, J. Jia, Z. Bu, B. Vinzamuri, A. Ramakrishna, K. Chang, V. Cevher, S. Liu, and M. Hong, “*BLUR: A Bi-Level Optimization Approach for LLM Unlearning*,” EACL, 2026.
- J. Jia, **H. Reisizadeh**, C. Fan, N. Baracaldo, M. Hong, and S. Liu, “*EPiC: Towards Lossless Speedup for Reasoning Training through Edge-Preserving CoT Condensation*,” Submitted, 2025.
- H. Reisizadeh**, B. Touri, and S. Mohajer, “*Almost Sure Convergence of Distributed Optimization with Imperfect Information Sharing*,” Accepted, Automatica, 2024.
- H. Reisizadeh**, B. Touri, and S. Mohajer, “*Distributed Optimization Over Time-Varying Graphs With Imperfect Sharing of Information*,” IEEE Transactions on Automatic Control, 2023.